

Project Repository and Documentation

- GitHub: <https://github.com/daniel-pgkf/s12-control-de-tono>
- Notion: https://app.notion.com/p/S12-Control-de-Tono-118f673dc6af8283b432819cb4957f24?source=copy_link

Project Description

Control de Tono is a project aimed at introducing core electronic engineering concepts through a hands-on, pedagogical approach for early-semester electronic engineering students. It uses audio systems as the case study through which students discover and learn to interpret tools such as Bode diagrams and filters.

Methodologically, the project centers on the design of a Frequency Response Analyzer (FRA) to generate visualizations of audio characteristics throughout the course, capable of analyzing the full audible range (20Hz-20KHz).

Hardware Architecture

- **Input chain:** 1/4” mono jack → high-impedance buffer (~1MΩ) with gain pot → mode switch (guitar / white noise) → ADC1 (96 kSps, I2S)
- **DUT slot:** 4-terminal modular connector for the network under test
- **Output chain:** ADC2 (96 kSps, synchronized) → output buffer + speaker driver → 1/4” mono jack
- **Microcontroller:** captures both ADCs via I2S, computes FFT of input/output, computes $H(f) = \text{FFT}(\text{output})/\text{FFT}(\text{input})$, controls the mode switch, footswitch, and 6 programmable buttons, sends data to the PC via USB
- **White noise generator:** excitation signal used to obtain the frequency response and Bode diagram (flat spectrum, allows measuring the full range in a single 500 ms to 1 s capture)

Bill of Materials

Exchange rate used for the COP item: 1 USD ≈ 3.445 COP (current TRM, subject to change).

Component	Function in the FRA	Qty.	Unit price	Subtotal	Link
1/4” stereo female PCB mount jack (DAIERTEK, 18pcs)	Input and output jacks	1	US\$13.99	US\$13.99	Amazon
ESP32-S3 N16R8 WROOM-1 (Hosyond, 3pack)	Microcontroller (project doc specifies ESP32-C3 Mini)	1	US\$18.99	US\$18.99	Amazon
PCM1808 24bit/105dB ADC (GODIYMODULES)	ADC1 and ADC2, 96 kSps I2S	4	US\$6.99	US\$27.96	Amazon
PCM5102A I2S DAC (GODIYMODULES, 2 units/listing)	Digital-to-analog conversion for monitoring/output	2	US\$8.88	US\$17.76	Amazon
NE5532P, dual low-noise op-amp	High-impedance input buffer / output buffer	1	US\$6.99	US\$6.99	Amazon
TLE2426CLP, rail splitter TO-92	Mid-supply bias point for analog stages on a single supply	1 (5 units/lot)	≈US\$7.74 (\$26.674 COP)	US\$7.74	AliExpress
3PDT footswitch + PCB + washer (5pcs)	Pedal/Bode mode switch, true bypass	1	US\$15.90	US\$15.90	Amazon
PAM8302, 2.5W class D amplifier (JESSINIE, 5 units)	Speaker driver in the output chain	1	US\$9.95	US\$9.95	Amazon
Tactile button kit 12x12x7.3mm, 8 colors (EGSCST, 120pcs)	6 programmable buttons	1	US\$8.99	US\$8.99	Amazon
Linear 50K potentiometer, single shaft (Ferwooh, 10pcs/lot)	Input gain pot (10K/50K/250K/500K for testing)	4	US\$6.99	US\$27.96	Amazon
Resistor kit, 25 values, 1000pcs (BOJACK)	Misc. passive components / test DUT networks	1	US\$9.99	US\$9.99	Amazon
Electrolytic capacitor kit, 24 values, 240pcs (ALLECIN)	Misc. passive components / test DUT networks	1	US\$9.99	US\$9.99	Amazon
170-point mini breadboard (CHANZON, 6pcs)	Prototyping of input/output chains	3	US\$7.99	US\$23.97	Amazon

Total: US\$200.18 (excluding shipping; shipping cost varies depending on whether orders are grouped under the same Amazon seller)